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Dramatic increases in redundant publications in the Generative AI era

Danny Maupin^{1†}, Tulsi Suchak^{1†}, Adrian Barnett² and Matt Spick^{1*}

Abstract

Background Redundant publication, the practice of submitting the same or substantially overlapping manuscripts multiple times, distorts the scientific record and wastes resources. Since 2022, publications using large open-science data resources have increased substantially, raising concerns that Generative AI (GenAI) may be facilitating the production of formulaic, redundant manuscripts. In this work, we aim to quantify the extent of redundant publication from a single, large health dataset and to investigate whether GenAI can create submissions that evade standard integrity checks.

Methods We conducted a systematic search for the years 2021 to 2025 (year to end-July) to identify redundant publications using the US Centers for Disease Control and Prevention National Health and Nutrition Examination Survey (NHANES) dataset. Redundancy was defined as publications analysing the same exposures associated with the same outcomes in the same national population. To test whether GenAI could facilitate creating these papers, we prompted large language models to write three synthetic manuscripts using redundant publications from our dataset as input, instructing them to maximise syntactic differences and evade plagiarism detectors. These three synthetic manuscripts were then tested using a leading plagiarism detection platform to assess their similarity scores.

Results Our search identified 411 redundant publications across 156 unique exposure–outcome pairings; for example, the association between oxidative balance score and chronic kidney disease using NHANES data was published six times in 1 year. In many instances, redundant articles appeared within the same journals. The three synthetic manuscripts created by GenAI to evade detection yielded overall similarity scores of 26%, 19%, and 14%, with individual similarity contributions below the typical 5% warning thresholds used by plagiarism detectors.

Conclusions The rapid growth in redundant publications (a 17-fold increase between 2022 and 2024) suggests a systemic failure of editorial checks. These papers distort meta-analyses and scientometric studies, waste scarce peer review resources, and pose a significant threat to the integrity of the scientific record. Current checks for redundant publications and plagiarism are no longer fit for purpose in the GenAI era; greater co-operation between publishers and modified guidelines will be needed to address new innovations in paper mill production.

Keywords NHANES, Integrity, COPE, Redundant publication, Paper mills, Generative AI, Plagiarism

[†]Danny Maupin and Tulsi Suchak contributed equally to this work.

*Correspondence:

Matt Spick
matt.spick@surrey.ac.uk

¹ School of Health Sciences, Faculty of Health and Medical Sciences, University of Surrey, Guildford, Surrey GU2 7XH, UK

² School of Public Health and Social Work, Queensland University of Technology, Kelvin Grove, Australia

Background

Artificial intelligence tools, including both large language models (LLMs) and easily deployable machine learning workflows, offer potential for both new insights and productivity gains to researchers [1]. One promising area is large and complex data resources, such as in healthcare and the biosciences [2, 3]. Inevitably, however, AI will multiply negative as well as positive outputs [4].



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The publishing industry has always had to contend with unethical behaviours, and guidelines on best practices in publishing are provided by the Committee on Publication Ethics (COPE). These negative behaviours include text and image recycling by authors [5]; plagiarism of others' work [6]; redundant submissions, with the latter including salami slicing or 'dividing data or research findings into their smallest publishable units to increase the author's publication output' [7, 8]; and systematic manipulation [9, 10] (Fig. 1).

We and others have previously documented the exploitation of Open Science data resources for the production of misleading research by entities such as paper mills (organisations that produce manuscripts for sale using derivative or fabricated text, images, or data) [11–13]. Hallmarks of these paper mill outputs include *p*-hacking (manipulating analyses or results reporting to obtain statistically significant *p* values) and data dredging (searching through data without any hypothesis for statistically significant relationships) [14, 15]. The ability of GenAI to synthesise new text in a handful of clicks poses an additional problem, as it bypasses both text recycling and plagiarism editorial checks, allowing mass manufacture of redundant manuscripts or simultaneous submission to journals. Whilst some publishers have processes to detect duplicate submissions [16], GenAI can assist with evading these checks. Entities such as paper mills might even be able to submit unlimited *conceptually identical* but *syntactically different* manuscripts, dramatically increasing the potential production rate of redundant

publications. When LLMs are combined with large public datasets such as NHANES or the FDA Adverse Events Reporting System (FAERS), the only rate limiting step remaining is the ability of publishers to review and accept manuscripts, placing ever more strain on editors and peer reviewers.

In addition to negative outcomes arising from the use of GenAI by bad actors such as paper mills, there may also be risks from cross-domain work that substitutes traditional human collaboration with AI. LLMs can act as expert summarisers, allowing a researcher in one field to rapidly acquire a functional understanding of another, thereby reducing the need for direct partnership with domain experts. For instance, a researcher looking for evidence to support a hypothesis might use a LLM to write statistical code to interrogate a large open access database. This approach streamlines the preliminary stages of interdisciplinary research, allowing for quick hypothesis generation and project scoping. This represents a clear productivity gain (which will be hard for small teams to pass up in a highly competitive landscape) [17], reducing the difficulties associated with assembling and managing cross-disciplinary teams (conflicting terminologies, differing methodological standards, logistical overheads) [18]. It can, however, also lead to epistemic trespassing, whereby researchers have a surface-level understanding but lack deep domain expertise. Whilst this has been previously described for systematic reviews and meta-analyses [19, 20], GenAI can enable such opportunistic authorship more broadly, allowing

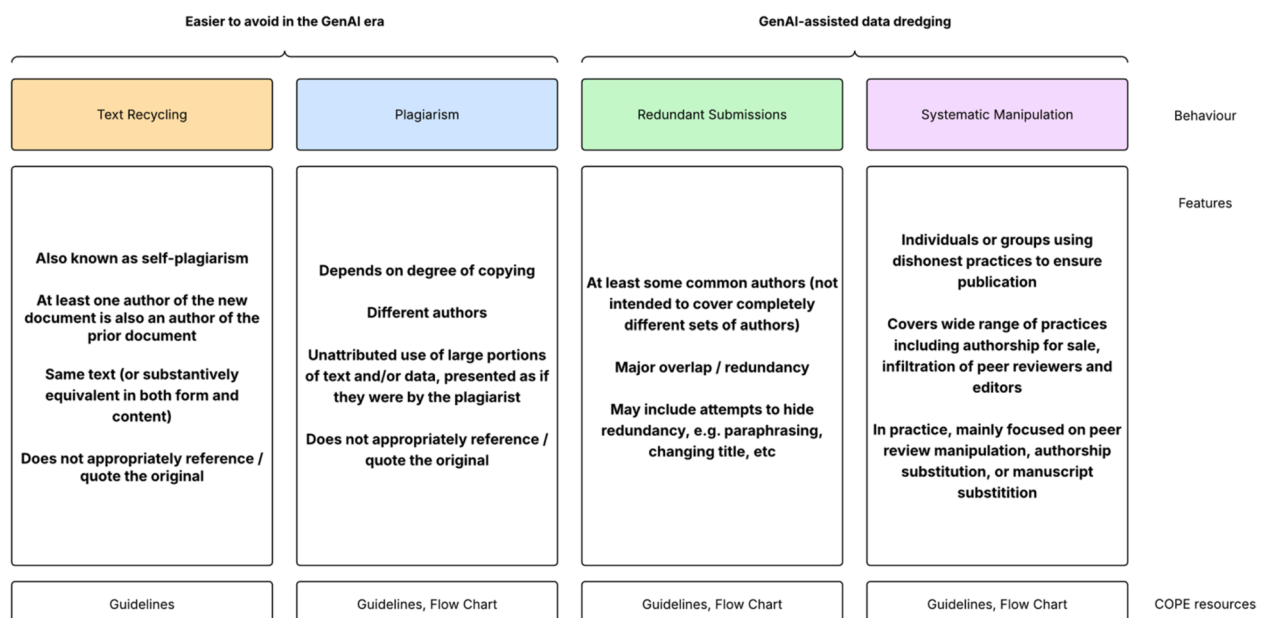


Fig. 1 Existing COPE guidelines around plagiarism and related examples of misconduct

researchers with limited domain expertise to efficiently scrape open datasets and find statistically significant relationships without the ability to put findings in their proper context.

Here, we aim to investigate whether GenAI is contributing to a breakdown in the effectiveness of integrity checks in the scientific publishing process. First, we examine whether redundant publication is becoming more prevalent, whether this is inadvertent epistemic trespassing (researchers with good intentions that have simply written papers where they lack the domain expertise to be aware of competitor groups and publications) or deliberate (through systematic manipulation). We use the NHANES dataset as our primary source [21] and investigate whether the incidence of redundant publications from this dataset has changed over time. Second, we investigate whether LLMs can facilitate the duplication of published works, by rewriting existing manuscripts and testing whether they pass plagiarism checks used by major publishers.

Methods

Literature search and manuscript matching

PubMed was used as the primary data source, with the following search string *‘(nhanes[tiab] OR national health and nutrition examination survey[tiab]) AND (“2021/01/01”[Date—Publication]: “2025/07/31”[Date—Publication])’*, which produced an initial list of 15,597 publications. Following this, articles referencing the distinct and independent Korean NHANES survey were excluded ($n=1657$). In addition, comments on publications were excluded ($n=50$). A list of exposures and outcomes was derived from our previous work on inappropriate manuscripts using the NHANES dataset [15], and this was added to based on the authors’ experience of more recent literature analysing biomarkers and predictors. In total, a list of 217 exposures/predictors and 162 outcomes was generated (Additional file 1: Table S1), including alternatives where appropriate.

Manuscript titles were searched for matching exposures and predictors by simple token matching, resulting in a set of 3465 manuscripts analysing the predefined list of simple exposures and outcomes. Sets of manuscripts with multiple papers analysing the same exposure and outcome were recorded. Prior to matching, where varying terms had been used for the same condition, these were manually replaced with the most common token (for example ‘depressive symptoms’ was converted to ‘depression’, as both outcomes are measured using the same NHANES questionnaire). The full list of original terms is provided in Table S1. Titles and abstracts were then manually inspected; studies were treated as effectively redundant publications if they analysed the same

exposure and outcome with only variations such as survey cohort included (e.g. 2014–2018 versus 2016–2020), male versus female, or minor variations in age ranges. This was done to prevent very similar analyses being treated as a separate research topic—for example two studies assessing infertility in women aged 18 to 45, or in women aged 20 to 45. More specific investigations, for example measuring associations between an exposure and an outcome in cancer survivors, were not included as redundant publications ($n=3054$), leaving 411 publications which under the definitions used in this work were redundant. This screening process was not intended to identify problematic publications in general (such as salami slicing or plagiarism); this workflow was intended to identify the specific issue of redundant/duplicate publication (Fig. 2).

Investigation of plagiarism detection

In order to pilot whether simple click-science workflows were sufficient to evade existing mainstream plagiarism detection, three ‘new’ manuscripts were generated from existing published papers. The exposure and outcomes chosen are set out in Table 1 alongside the PMIDs of the ‘source’ publications for the synthesis.

These exposure/outcome pairings were selected as examples due to the high number of duplicates found during pilot searches, and the large amount of available literature on the topics, which represent sources of considerable public interest. First, ‘new’ manuscripts were generated by three of the authors (D.M., T.S., and M.S.) independently using LLMs (GPT-4, GPT-5, and Google Gemini 2.5 Pro were used as easily accessible tools that might be employed by unethical actors) to rewrite existing manuscripts, with the only requirement that prompts explicitly specify that the new text should be syntactically different to the original text to avoid plagiarism detection. In addition to the main sections, LLMs were used to generate new syntactically different table titles, figure legends, and end disclosures. Second, the ‘investigated’ cohort was changed, and all numbers were adjusted either by instructing an LLM to create new tables, or by manual adjustments to reflect the changes that subtle alterations to NHANES cohorts can produce. Third, all text was manually adjusted where appropriate to reflect the numerical changes, to ensure that text referred to the correct figure and table numbers, and to remove repetition of acronyms (for example spelling out LE8 as ‘Life’s Essential 8 (LE8)’ in full each time mentioned) and formatted. It should be noted that the LLMs could not be prompted to produce completely error-free manuscripts, but the overall production process in each case was around 2 h per paper.

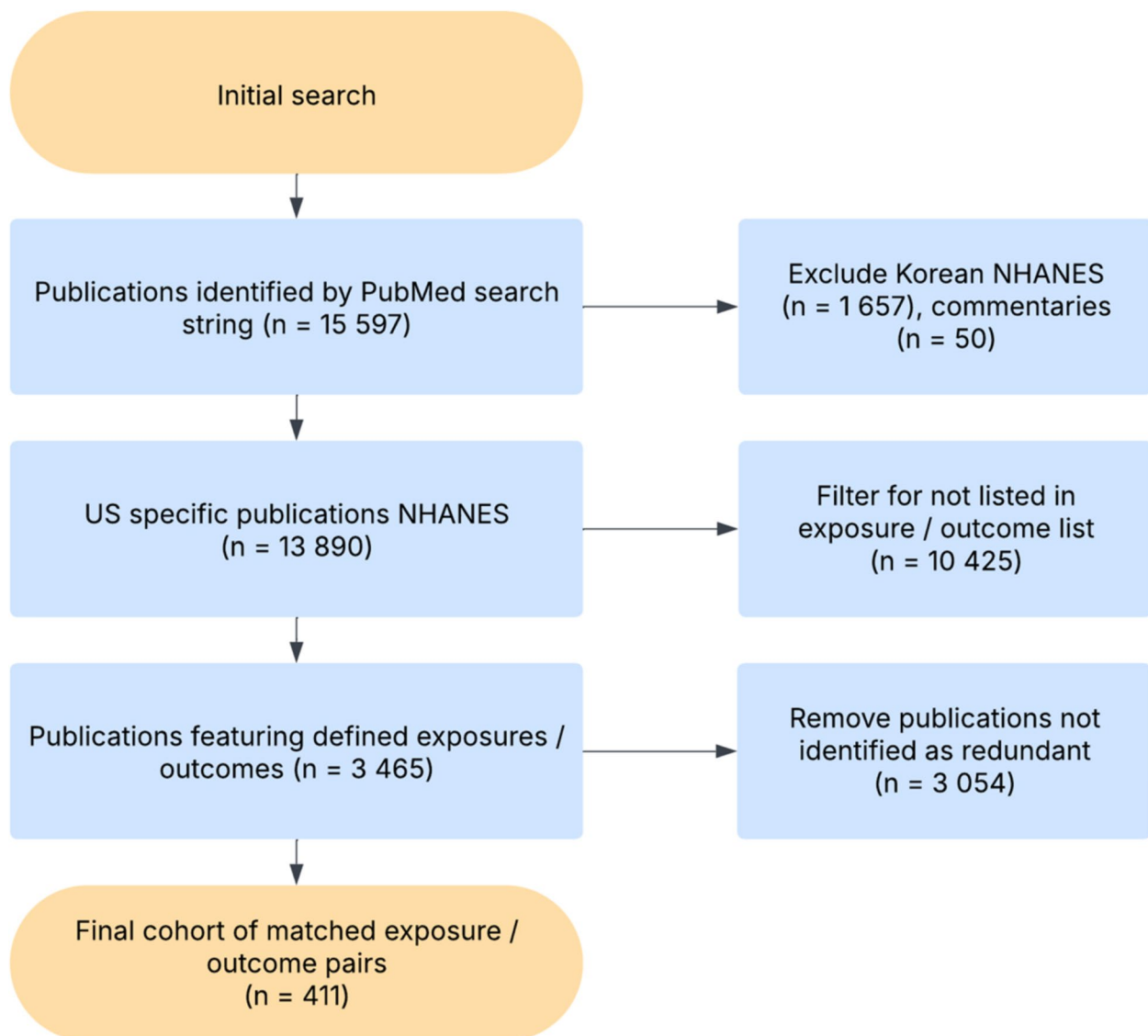


Fig. 2 Flowchart of inclusion and exclusion criteria for redundant publication search

Table 1 Pairings of exposures and outcomes selected for the production of synthetic manuscripts

Exposure–outcome pair	LE8 and periodontitis	BRI and infertility	Oxidative balance score and chronic kidney disease
Existing number of papers	6	5	6
PMIDs of existing papers	38964852	39545053	38919480
	38681053	39742100	39830066
	38178120	39696159	39399530
	39739130	39827325	38877058
	38622031	39959618	39752406
	39730816		38725577

LE8 Life's Essential 8, BRI Body Roundness Index, PMID PubMed unique identifier.

Subsequently, the three generated documents were submitted to iThenticate to assess whether a major current plagiarism detector would identify them as unoriginal documents [22]. We followed guidance from Springer Nature's documentation on Crossref Similarity Check/iThenticate that it is 'more important to look at the individual scores of the sources than the overall similarity index' and that individual scores in the 1% to 5% range offer 'in general no sign of potential plagiarism' [23]. Therefore, we treated individual scores over 5% as generating potential red flags for editors and reviewers.

Results

Literature search and manuscript matching

The literature search and manuscript matching process generated 411 paired redundant exposure–outcome documents. The most common result was two papers covering the same exposure/outcome pairing ($n=190$ publications). The largest number of redundant analyses was six, occurring for three pairings ($n=18$ publications). The 20 most frequent exposure–outcome pairs and the distribution of redundancies are shown in Fig. 3A

and B respectively. Figure 3C illustrates the incidence of the duplicates identified over time; there were very few redundant publications relating to NHANES prior to 2023 (12 in 2022 and 3 in 2021), with the 2024 count of 198 representing a 17-fold increase in 2 years. In comparison, the number of papers per year using the NHANES dataset prior to 2023 was 2535 in 2022, 1944 in 2021, and 5182 in 2024, a twofold increase in 2 years.

The 411 articles were published by 112 journals (Fig. 4A), with 44% concentrated in just five journals. The most common statistical method used in the duplicated research publications was logistic regression across multiple models adding covariates cumulatively, followed by linear regression, with a small number of other methods used (Fig. 4B).

Submission and publication dates

Three illustrative pairings were chosen from those with the highest rates of redundant publications for the generation of synthetic manuscripts. These were chosen as they should be harder to be duplicate and evade plagiarism detection (as more redundant instances are available

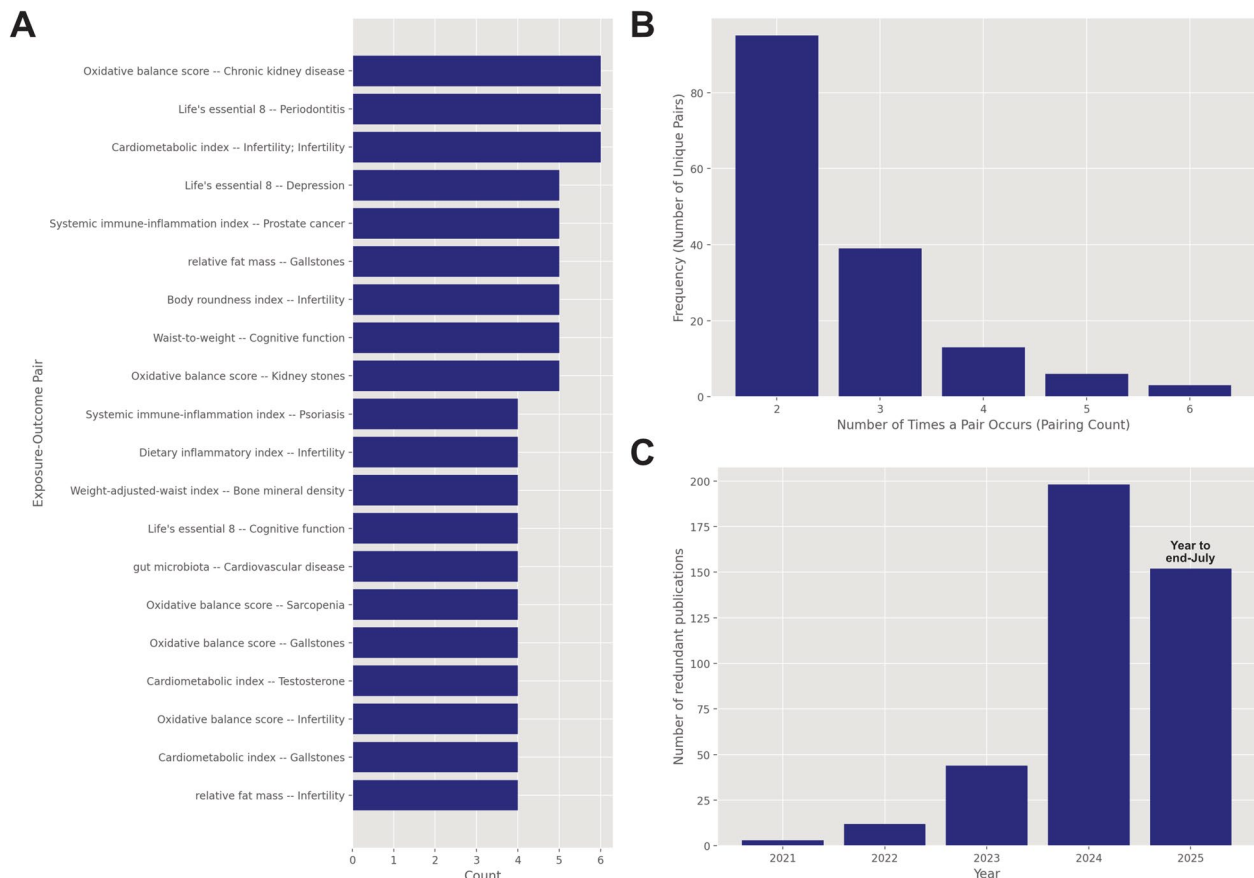


Fig. 3 **A** 20 most frequent exposure–outcome pairs, **B** distribution of redundant publications, and **C** time trend of redundant publications, January 2025 to July 2025

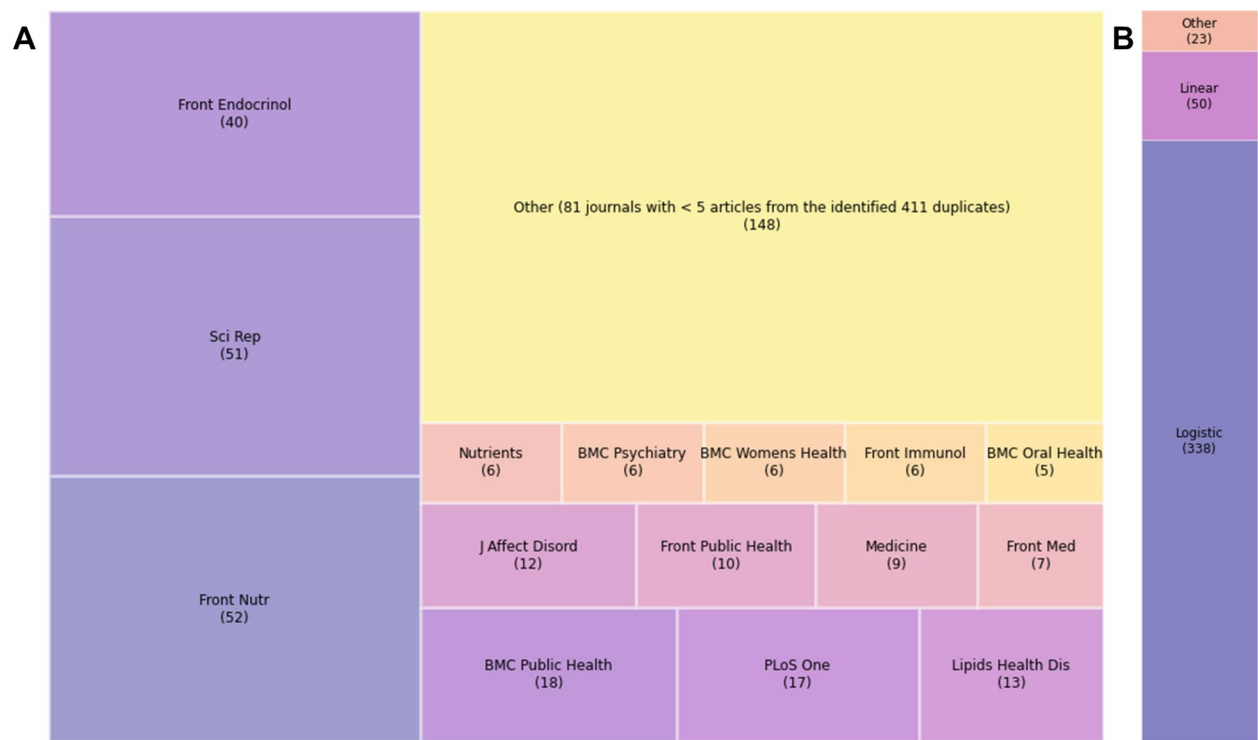


Fig. 4 **A** Tree map of journals using the number of redundant papers and **B** breakdown of statistical methods used in the redundant papers. We use the abbreviated journal name—full journal names are provided in Additional file 1: Table S2

in the public domain). Submission and publication dates for these redundant publications are shown in Fig. 5, together with the journals that published them. Articles commonly followed a pattern of simultaneous or near-simultaneous submission, but also included examples of submissions after the publication of previous works. In some cases, the same journals published the redundant manuscripts. This occurred 67 times across the cohort of 411 publications, with *Frontiers in Nutrition* publishing the same redundant pairing on 29 occasions, and *Scientific Reports* on 10 occasions (the full list of journals is included in Additional file 1: Table S2).

Evasion of plagiarism checks

The three synthetic manuscripts generated by LLM were then tested using iThenticate. The results are shown in Table 2 for the total similarity scores and for the five main individual similarity sources for each manuscript. The total scores including bibliographical matching are included for reference. All three manuscripts had percentage matching scores that were below 30% overall and no individual source percentages exceeded 5%, and so would not have raised automated red flags in editorial pipelines [23].

The LLM-rewritten manuscripts are presented in Additional file 2 with watermarking and disclaimers. The

watermarks and disclaimers were not included in the versions uploaded to iThenticate.

Discussion

Plagiarism and related misconduct such as submissions of redundant manuscripts have been a constant in scientific publishing, and many checks and balances exist to guard against these problems [24, 25]. The results presented here, however, are suggestive that these checks are not adequate in the GenAI era, given the sharp acceleration since 2022 in the number of redundant publications (2024 exhibiting a 17-fold change versus 2022, for example). Our workflow identified 411 total redundant publications from a single data source, NHANES, but it should be noted that the matching procedure was heavily automated (and targeted at exposures and outcomes that have previously been associated with paper mill activity), and better-concealed duplications would have escaped this process (a full manual text check of 15,000+ manuscripts being beyond the scope of this work). This has happened at the same time as wider growth in submissions putting editorial and peer reviewing processes under increased strain [26, 27], possibly increasing the dependence on automated systems such as iThenticate. The workflow also identified some journals as accepting a disproportionate number of redundant publications, which may

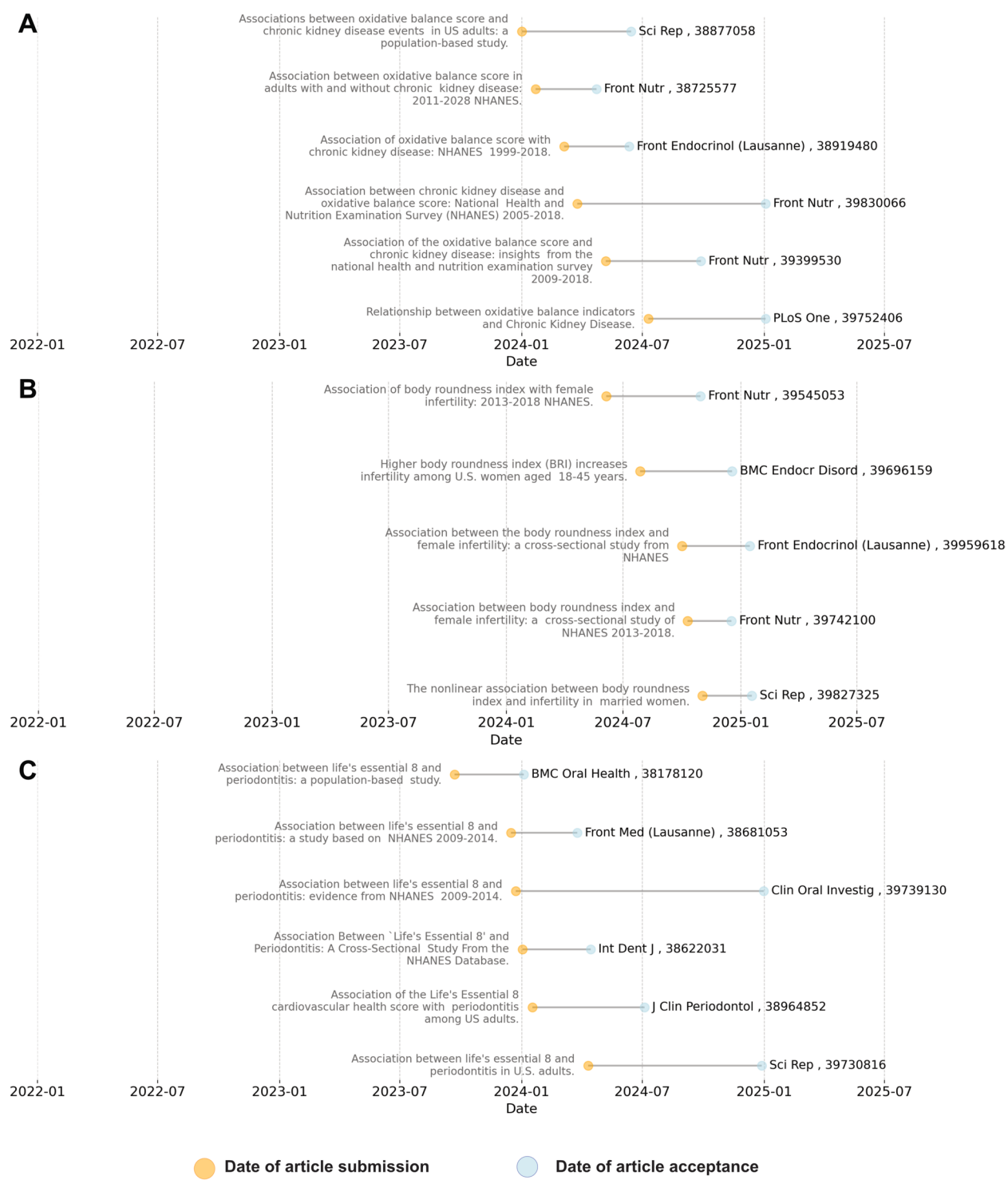


Fig. 5 Submission and publication date (year-month) plots for three outcome–exposure pairings, with publication titles to the left of the submission dates (orange dots) and journal title and publication PMIDs to the right of acceptance dates (blue dots): **A** oxidative balance score and chronic kidney disease, **B** Body Roundness Index and female infertility, and **C** Life's Essential 8 and periodontitis

Table 2 Summary of iThenticate checks for three synthetic manuscripts, processed on 26 August 2025

Exposure–outcome pair	LE8 and periodontitis	BRI and infertility	Oxidative balance score and chronic kidney disease
LLM used	ChatGPT 5	Gemini 2.5 Pro	ChatGPT 4o
References updated	No	Yes	No
Tables by LLM or manual	LLM	Manual	Manual
Total score excluding bibliography (%)	26%	19%	14%
#1 Source (%)	frontiersin.org (5%)	frontiersin.org (4%)	frontiersin.org (2%)
#2 Source (%)	medrxiv.org (2%)	medrxiv.org (1%)	crossref source (1%)
#3 Source (%)	ncbi.nlm.nih.gov (2%)	ncbi.nlm.nih.gov (1%)	mdpi.com (1%)
#4 Source (%)	crossref source (1%)	crossref source (1%)	ncbi.nlm.nih.gov (1%)
#5 Source (%)	multicentredatabase.net (1%)	researchgate.net (1%)	researchgate.net (1%)
Total score including bibliography (%)	37%	21%	25%

LE8 Life's Essential 8, BRI Body Roundness Index.

reflect paper mills targeting journals where fabricated manuscripts have already been accepted [28], perhaps reflecting perceived weaknesses in their editorial checks.

Our findings on duplicates illustrate the scale of the problem (of the 3465 articles analysing our target exposures and outcomes, 12% were redundant publications), but do not provide the mechanism by which it is happening. Here, we have demonstrated that ‘new’ syntactically different manuscripts can easily be generated that avoid text recycling or textual plagiarism. Whilst our work does not represent a systematic investigation of how easily plagiarism checks can be avoided, the three pairings chosen here did not exceed warning thresholds for individual source similarity scores of 5%, and were also free of ‘tortured phrases’, another indicator often used to detect problematic publications [29]. Interestingly, for all three synthetic manuscripts the Frontiers family of journals provided the highest ‘individual source similarity’ scores, which likely reflects Frontiers accepting more of these types of publications than other journals (Frontiers accepted 29% of the redundant publications identified in this work). Our synthetic manuscripts also passed thresholds despite our selecting highly duplicated pairings of exposures and outcomes, and the substantial time gap between publication of our ‘source documents’ and our submitting of the plagiarised manuscripts to iThenticate.

Redundant publications can also amplify bad research. For example, vitamin D intake can be correlated with depression, but using NHANES data the association between insufficient serum vitamin D and depression has been shown by Diaz-Amaya et al. to be fully attenuated after adjusting for food security and diet quality [30]. Nonetheless, here we have identified three publications taking a simplistic single-factor approach reporting that

serum vitamin D is associated with depression, findings which we regard as unsound. Therefore, misleading publications on serum vitamin D using NHANES outnumber the more thorough analysis in a 3:1 ratio. A simple comparison by Google Scholar shows citations for these publications in a 30:2 ratio. This is likely to be a major problem for those that are not domain experts—many readers are likely to take the ‘weight’ of publication or citation counts as indicative of evidence. This can then further propagate low-quality results.

These findings extend previous work highlighting mass-manufactured low-quality research generated from publicly accessible datasets, leading to the literature being oversaturated with repetitive and potentially misleading findings [15, 31]. Combined with the strong incentive to publish frequently (culturally for authors and financially for journals), this has created a system where quantity is able to displace quality. There is growing evidence that these features have been exploited by ‘paper mills’, organisations that will produce manuscripts on a large scale, typically with minimal originality, and offer authorship slots to academic customers to meet the demand for rapid publication [32, 33]. NHANES is an attractive target for these organisations: it is publicly available, has no registration or reporting requirements, and can be mined repeatedly with surface-level variations in study design. It is, however, important to recognise that the methods described in this work will not just be applicable to NHANES, albeit they may be easier to identify for this dataset due to the sheer volume of manuscripts. There is no reason why redundant submissions would not be enabled by GenAI for other health datasets, or for other types of research, such as systematic review papers, for example. We see risks that paper mills may migrate into these areas if negative attention around data

resources such as NHANES limits publication opportunities. Epistemic trespassing is also likely to see growth, as researchers new to an area familiarise themselves with a topic using GenAI and see that it offers an easier route to publication.

Addressing these challenges will not be straightforward. Whilst some publishers have taken action on the exploitation of Open Science datasets [34, 35], this may have diverted the flow of low-quality submissions rather than reduced it [36, 37]. COPE definitions do not fully capture this type of misbehaviour, as their categorisation of ‘redundant publication’ deals with the same authors submitting to multiple journals [8]. We contend that this definition does not reflect the reality that the true author is concealed by contract cheating, whereby the disclosed authors appear to be different (but have in fact purchased authorship from the same source). Proving this in individual cases is not easy, but we see no other plausible explanation for the aggregated surge in redundant publications described here.

Limitations

Whilst the overall trends are informative and illustrate a growing problem, a number of limitations should be stressed. First, from reviewing manuscripts alone, it is not possible to differentiate between epistemic trespassers that are using GenAI to produce outputs in areas where they lack expertise, versus unethical actors such as paper mills deliberately using strategies such as simultaneous submission to multiple venues. It should also be noted that sets of duplicate publications do not automatically mean problematic behaviour, for example failure to notice that a similar paper has been already published, or for the first paper in each ‘set’. Second, here we took a conservative approach to minimise false positives, by excluding associative studies that analysed specific subsets of data. This may understate the scale of the problem. Third, this work has identified the potential for using LLMs to quickly rewrite manuscripts. Here, however, we have shown illustrative examples. This is not a comprehensive analysis across a large number of manuscripts to identify how often tools such as iThenticate can be ‘fooled’. The evidence from the overall publication of duplicated topics, however, suggests that it is unlikely that iThenticate and similar tools are well prepared for the changing landscape, and it is the nature of LLMs that should a submission fail, the LLM can simply reword the document again until it passes. Fourth, interventions were required to correct simple errors introduced, and this typically required an additional 2 h of manual checking. The prompts used here were not able to produce new error-free manuscripts in a single click, albeit the reduction in time would in our

view represent a massive time saving. Fifth, our workflow was designed to test whether documents rewritten by LLMs would pass automated checks. We do not suggest that our documents would pass the desk editor or peer review checks in place at responsible journals, and further interventions and editing would be required to do so. Many journals are investing in customised solutions, and we would stress that we welcome any opportunities to collaborate with publishers and vendors to identify whether our synthetic manuscripts would pass more advanced checks.

Conclusions

These results demonstrate that current checks for redundant publications and plagiarism are no longer fit for purpose in the GenAI era, with a sharp acceleration coinciding with the initial launch of ChatGPT in November 2022. There are, however, few easy solutions. We expect greater emphasis on AI and other forms of detection from publishers as another ‘automated’ check to reinforce desk rejection, especially as publication volumes appear to be expanding beyond the ability of manual checks and balances such as peer review to cope. For known ‘templates’ such as NHANES, FAERS, or two-sample Mendelian randomisation (e.g. derived from FinnGen or the UK Biobank), we believe publishers need to be especially vigilant for formulaic manuscripts. In the long-run, greater co-operation between publishers for a register of submissions in these areas might help to reduce redundancy, and the International Association of Scientific, Technical & Medical Publishers (STM) is also trying to take action in this area [38]. We believe that updated/modified guidelines will be required from COPE to address these new innovations in paper mill production, especially to move away from current narrow definitions of redundant publication that focus on visible author overlap, for example. Without action to address these new GenAI related challenges, however, we expect the problem to continue to grow, making scientometric and meta-analyses more challenging, wasting publisher and peer reviewer resources, and reducing the integrity of the scientific record.

Abbreviations

LE8	Life's Essential 8
BRI	Body Roundness Index
PMID	PubMed unique identifier
GenAI	Generative AI
NHANES	National Health and Nutrition Examination Survey
COPE	Committee on Publication Ethics
LLMs	Large language models
FAERS	FDA Adverse Events Reporting System
STM	International Association of Scientific, Technical & Medical Publishers

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-025-04569-y>.

Additional file 1. Table S1 List of exposures and outcomes used to identify redundant publications. Table S2 Complete dataset of exposure versus outcome pairings including PMIDs.

Additional file 2. Example prompts used to synthesise manuscripts, together with watermarked copies of the three synthetic manuscripts used to test plagiarism detectors.

Acknowledgements

The authors wish to acknowledge the work of the United2Act network in its work to highlight the threats to scientific integrity from paper mill activity [39], in addition to the Collection of Open Science Integrity Guides (COSIG) [40]. The authors also wish to declare that GenAI (large language models) was used to prepare the synthetic manuscripts in this work, as described under Methods.

Authors' contributions

MS and AB designed the study, developed the methodology, provided resources, and supervised the project. TS, MS and DM prepared the synthetic manuscripts, provided analysis of publication data and jointly produced the original draft; MS created the visualizations. MS and AB reviewed and edited the manuscript. All authors read and approved the final manuscript. MS is the guarantor of this manuscript and is responsible for the overall content. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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Data availability

The data underpinning this work can be reproduced from the PubMed database at <https://pubmed.ncbi.nlm.nih.gov/>. The complete dataset of exposure versus outcome pairings including PMIDs for the 411 redundant publications included in this work is included in Additional File 1, Table S2.

Declarations

Ethics approval and consent to participate

No patients or members of the public were directly involved in this study as no primary data were collected.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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